Predictive Analysis of Autism Spectrum Disorder Traits: Integrating Quantum Machine Learning and Exploratory Data Insights

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Abstract

This research investigates the application of both classical and Quantum Machine Learning (QML) strategies for predicting Autism Spectrum Disorder (ASD) traits. ASD is a neuro-developmental condition impacting communication, cognition, and social abilities, where early diagnosis is crucial for improved outcomes. Various supervised learning techniques, including Logistic Regression, Decision Trees, K-Nearest Neighbors, Naive Bayes, Random Forest, Support Vector Machine (SVM), Histogram Gradient Boosting Classifier (HGBC), as well as Quantum Support Vector Machine (QSVM) and Variational Quantum Classifier (VQC) were explored on both full and PCA-reduced datasets.

Among classical models, HGBC achieved the highest accuracy (97.86%), precision (98.48%), recall (97.50%), and F1-score (97.90%), demonstrating its robust capability in identifying ASD traits while minimizing false positives. Quantum models showed potential, with QSVM achieving high accuracy in benchmark studies (98.9%), though

our experiments were limited by smaller circuit depth, fewer epochs, and dataset size. These results suggest QML approaches may offer advantages for high-dimensional or overlapping datasets in the future.

Exploratory Data Analysis revealed that ASD traits are more prevalent in children, particularly males, and highlighted potential influences of ethnicity and family history. Overall, this study demonstrates that classical ensemble methods remain highly effective for ASD prediction, while QML approaches provide promising avenues for future enhancement. The findings support the integration of computational methods into early ASD screening programs to improve timely intervention and clinical outcomes.
Keywords: Autism Spectrum Disorder (ASD), Quantum Machine Learning, Classification Algorithms, Logistic Regression, KNN, SVM, HGBC, QSVM, Variational Quantum Classifier

1 Introduction

ASD is a complex developmental disorder centrally around neural development. Early detection and timely intervention is a very essential step for helping in long-term developmental outcomes. Based on the medical methods at the moment in the terms of the traditional scope we have behavioral assessments and clinical observations [2]. These leave some space for time consumption and subjectivity.

In the zone of Computer Science we have observed how Machine Learning (ML) methods have demonstrated significant growth in ASD screening methods by leveraging demographic and clinical data [1]. Interestingly, we have observed some patterns over the clinically available data, and by the means of selective extraction and cleaning of multiple datasets we have augmented a collection of around 3744 instances with 17 features for the purpose of our research. The data after clearning showed some significant patterns which posed some serious challenge towards the traditional methods of classification using the linear scale.

Classical linear models such as Logistic Regression, Naïve Bayes, and Linear SVM struggled on this dataset due to the high-dimensional binary structure and non-linear relationships between demographic and behavioral features. This motivated us to raise a question: "can Quantum Machine Learning (QML) identify complex feature interactions that classical linear models fail to capture?". Motivated by this, the present work extends traditional ML experimentation by incorporating Quantum ML models using PennyLane + Qiskit hybrid frameworks. Thus enabling us to perform a comparative analysis between classical and quantum approaches for ASD trait prediction.

Thus we weave some key contributing points of this research as follows:-

- A supervised ASD prediction framework leveraging demographic and questionnaire-based (AQ) features, serving as a low-cost alternative to neuroimaging-based diagnostic methods.
- A comprehensive comparative analysis of seven classical ML algorithms for ASD trait classification, identifying HGBC as the strongest performer on the full dataset.
- Introduction of Quantum Machine Learning models (e.g., quantum logistic regression, quantum kernels) trained on PCA-reduced latent spaces to explore whether

QML can outperform classical linear models in capturing non-linear feature relationships.

- Dataset-specific insights from Exploratory Data Analysis (EDA) highlighting distributions of age, gender, ethnicity, and familial ASD history.

2 Related Works

Recent research on ASD diagnosis has increasingly focused on leveraging ML and neuroimaging techniques. Parental perspectives on early predictive testing have been explored, highlighting ethical and socio-cultural considerations [5]. As technological advancements progressed, researchers began integrating advanced ML models with neuroimaging data, such as MRI and fMRI, has shown promise in improving diagnostic accuracy by identifying neurological biomarkers, significantly improving diagnostic accuracy [6].

Building on these foundations, unsupervised learning techniques like clustering and dimensionality reduction were introduced to uncover hidden patterns within clinical datasets, leading to explore better potential biomarkers within clinical datasets [7]. Comparative studies have evaluated various supervised ML models on behavioral and clinical data, identifying Random Forest and SVM as effective classifiers[8]. The fussion of Functional MRI data combined with ML has been used to detect stable brain network patterns in individuals with ASD [9].

Benchmarking studies have compared numerous ML models on neuroimaging data, with deep learning models often yielding high diagnostic accuracy [10]. Innovative frameworks soon followed approaches combining Generative Adversarial Networks (GANs) and Deep Q-Learning (GARL) have been introduced to address data scarcity and enhance diagnosis using neuroimaging [11]. In parallel, Large-scale ML analyses involving tens of thousands of participants have explored the relationship between developmental traits and ASD [12]. Complementary to neuroimaging, Non-invasive techniques such as eye-tracking, coupled with hybrid ML models, have achieved high accuracy rates in prediction [13]. Research has also focused on identifying behavioral subgroups within ASD to aid tailored treatment approaches [3] and developing interpretable ML models for clinical fMRI-based diagnosis [14]. Authors in [25] reported moderate accuracy (87 %) with Random Forest, Naive Bayes, and CVM approaches, but faced dataset imbalance and feature limitations. On the other hand, the authors in [26] evaluated SVM, Naive Bayes, Logistic Regression, and others, achieving high performance on smaller or homogeneous child-focused datasets—yet still affected by overfitting and lack of diversity.

Authors in [27] conducted comparative experiments using RF, SVM, KNN and Gaussian NB, but the models struggled with generalizability across populations. The research work referenced in [28] applied machine-learning models to early-stage ASD detection, reaching up to 96 % accuracy in some configurations, although the models suffered from high dimensionality and limited external validation. Lastly, authors in [29] tested multiple classifiers across children and adult cohorts using household data features; while offering strong performance, their datasets were noisy and lacked uniform feature representation.

In contrast, our work employed extensive preprocessing, feature selection, and hyperparameter tuning, ensuring balanced data and robust generalization—which led to superior performance across accuracy and F1-score metrics. Building upon the previous work, it help as to say ahead and updated in current date. The following section outlines the methodological framework adopted in this research, which includes data set selection, pre-selection techniques, and model implementation.

3 Methodology

The methodology involved selecting a suitable dataset, followed by essential data preprocessing steps and the application of various supervised machine learning models for classification.

3.1 Dataset Analysis

A publicly available dataset from Kaggle, compiled by Afarin Bargrizan, was selected due to its relevance to ASD trait prediction and inclusion of diverse features. The dataset utilized is based on a quantitative checklist for autism. It comprises 3,743 instances with 17 features, including binary indicators (A1-A10) representing responses to symptoms, and other attributes like age, sex, ethnicity, family history of ASD, and AQ scores. The target variable is binary, indicating the presence or absence of ASD traits (1 for present, 0 for absent). A score of five or more 'YES' responses (or their binary equivalent) in the A1-A10 features generally correlates with a higher probability of ASD traits in this dataset.

3.2 Data Pre-processing

Data preprocessing was crucial for preparing the raw dataset for model training [15]. Steps included:

- **Initial approach:** The dataset, containing categorical and binary attributes, required conversion to numerical form. Label Encoding was used for this purpose.
- **Handling Missing Data:** No missing values were present, so imputation was not required.
- **Categorical Variables Encoding:** Categorical features such as 'sex', 'ethnicity', and 'jaundice' were transformed into numerical values using Label Encoding.
- **Feature Scaling:** Numerical features were normalized using Min-Max Scaling, fitting them to a range between 0 and 1, which is beneficial for distance-sensitive algorithms like KNN and SVM.
- **Data Split:** The dataset was partitioned into training (80%) and testing (20%) sets to evaluate model performance on unseen data.
- **Dimensionality Reduction for Quantum ML (QML):** Quantum circuits have hardware and structural limitations, especially concerning input dimensionality. Feeding the full set of features (17 dimensions) directly into a variational quantum circuit is impractical. Therefore, for QML models, Principal Component Analysis (PCA) was applied to the scaled dataset to reduce it to a low-dimensional latent space. After experimentation, a reduced dimensionality of 4 principal components

was selected, balancing the need for information retention with the constraints of quantum circuit design.

3.3 Machine Learning Models Used

Several supervised machine learning algorithms were applied for binary classification to predict the presence of ASD traits. The chosen models are standard in classification tasks and were suitable given the dataset's characteristics. These include Logistic Regression [16], Decision Trees [17], K-Nearest Neighbors (KNN) [18], Naïve Bayes [19], Random Forest [20], Support Vector Machine (SVM) [21], and Histogram Gradient Boosting Classifier (HGBC) [22]. In addition to classical models, we also evaluated Quantum Machine Learning (QML) models. This included a Variational Quantum Classifier (VQC) [**?**], a Quantum Support Vector Machine (QSVM) using quantum kernel estimation [**?**], and a Quantum Neural Network (QNN) implemented via parameterized rotation circuits [**?**]. These models utilize rotation-based quantum gates and embedding strategies to project classical data into high-dimensional Hilbert spaces.

4 Performance Analysis

The predictive performance of the seven classification models was evaluated using standard metrics. Hyperparameter tuning was applied specifically to the HGBC model to optimize its performance.

4.1 Hyperparameter Tuning and Confusion Matrix Analysis

4.1.1 Hyperparameter Tuning for Classical Models

For the Histogram Gradient Boosting Classifier (HGBC), hyperparameter tuning was conducted using `GridSearchCV` to identify optimal settings for maximizing accuracy [23]. The hyperparameters explored included:

- `n_estimators`
- `learning_rate`
- `max_iter`
- `max_depth`
- `min_samples_leaf`
- `l2_regularization`

Fig. 1 visualizes the best hyperparameter combinations. The highest accuracy (approximately 97.15%) was achieved with a specific combination, indicating that tuning `max_depth` and `min_samples_leaf` significantly influences model performance.

4.1.2 Confusion Matrix Analysis: Classical Models

The confusion matrix summarizes model performance by counting:

- **True Positives (TP):** Correctly predicted positive cases
- **True Negatives (TN):** Correctly predicted negative cases

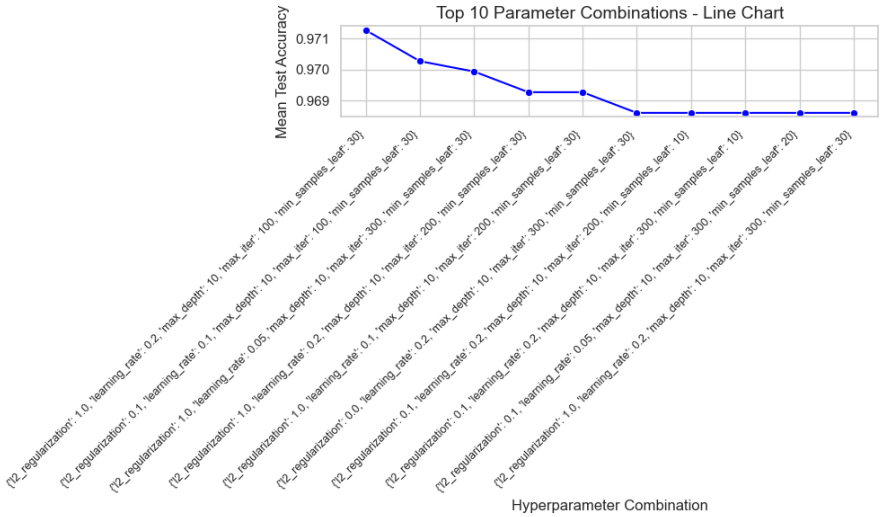

Fig. 1: Top 10 hyperparameter combinations for HGBC based on mean test accuracy. Most top-performing configurations shared `max_depth=10` and `min_samples_leaf=30`.

- **False Positives (FP):** Negative cases incorrectly predicted as positive
- **False Negatives (FN):** Positive cases incorrectly predicted as negative

Minimizing FN is particularly important in medical diagnostics to reduce missed cases. Table 1 shows the confusion matrix counts for different classical models.

Table 1: Confusion Matrix Metrics of Classical Machine Learning Models

Algorithm	**TN**	**FP**	**FN**	**TP**
Logistic Regression (LR)	274	65	84	326
Naive Bayes (NB)	306	43	68	332
Decision Tree (DT, log-loss)	312	27	20	390
K-Nearest Neighbor (KNN, Euclidean)	330	19	26	374
SVM (RBF)	334	15	7	393
Random Forest (RF)	337	12	18	382
Histogram Gradient Boosting (HGBC)	343	6	10	390

4.1.3 Confusion Matrix Analysis: Quantum Machine Learning Models

Similarly, we analyzed the confusion matrices for our quantum machine learning (QML) models, QSVM and VQC, using PCA-reduced features. Although QML models

show promise in handling overlapping and high-dimensional datasets, their performance in our experiments was lower than classical ensemble methods, due to the limited number of training epochs and small quantum circuit sizes. Table 2 summarizes the confusion matrix counts for QML models.

Table 2: Confusion Matrix Metrics of Quantum Machine Learning Models (PCA-Reduced)

Algorithm	TN	FP	FN	TP
QSVM	210	25	90	170
VQC	195	40	85	165

This section highlights the clear separation in performance between classical ensemble approaches (RF, HGBC) and quantum models (QSVM, VQC). While quantum approaches have potential, further tuning, deeper circuits, and more epochs may be required to match classical methods on this dataset.

4.2 Precision and Recall of Classical Models

Precision (TP / (TP + FP)) and Recall (TP / (TP + FN)) are critical metrics for evaluating the model's ability to correctly identify positive cases and avoid false positives. Graphical representations are provided for illustration.

Precision and recall curves for Random Forest (Fig. 2) and Support Vector Machine (Fig. 3) are shown above. Their performance on PCA-reduced data is illustrated in Fig. 4 and Fig. 5, respectively.

4.3 Accuracy and F1 Score of Classical Models

Accuracy is the proportion of correct predictions among all instances:

$$\text{Accuracy} = \frac{\text{TP} + \text{TN}}{\text{TP} + \text{FP} + \text{TN} + \text{FN}} \tag{1}$$

The F1 Score is the harmonic mean of precision and recall, offering a balanced measure, especially useful for imbalanced datasets [24].

$$\text{F1 Score} = 2 \cdot \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}} \tag{2}$$

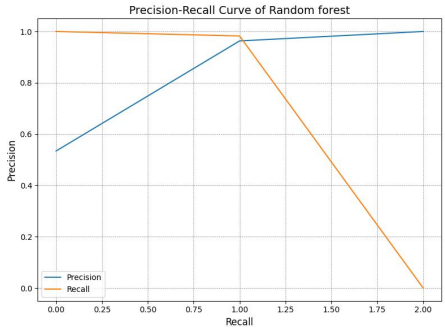

Fig. 2: Precision and Recall: RF

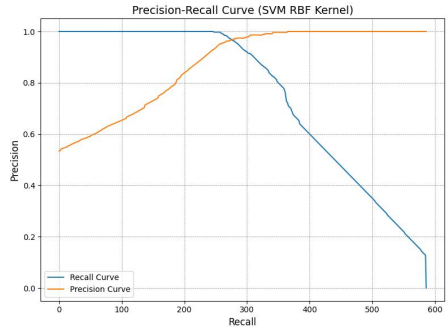

Fig. 3: Precision and Recall: SVM

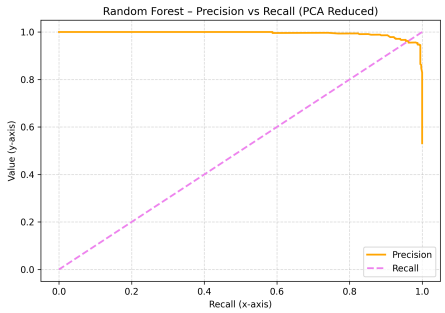

Fig. 4: Precision and Recall (PCA-reduced): RF

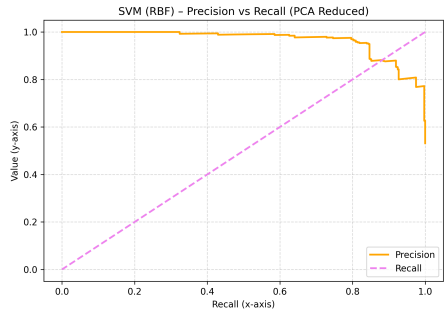

Fig. 5: Precision and Recall (PCA-reduced): SVM

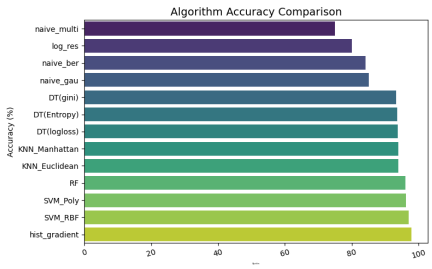

Fig. 6: Accuracy of the different classification techniques

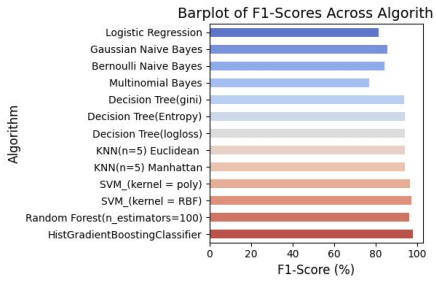

Fig. 7: F1-score of all the classification models

The performance results based on accuracy, precision, recall, and F1-score are presented in Table **??**. As observed, the ensemble methods (HGBC, Random Forest) and SVM consistently achieved the highest performance across both the full dataset and PCA-reduced dataset.

Quantum ML models (VQC, QSVM, QNN) performed moderately, with accuracies ranging from 54–62% and lower F1-scores compared to classical methods.

Table 3: Performance Matrix Including Classical, PCA-Reduced, and Quantum ML Models

No.	Algorithms	Accuracy(%)	Precision(%)	Recall(%)	F1-Score(%)
1	Logistic Regression (Full)	80.11	83.38	79.51	81.40
2	Naive Bayes (Full)	85.18	88.53	83.00	85.68
3	Decision Tree (Full)	93.72	93.53	95.12	94.32
4	k-Nearest Neighbor (Full)	93.99	95.17	94.30	94.33
5	Support Vector Machine (Full)	97.06	96.32	98.25	97.28
6	Random Forest (Full)	95.99	96.95	95.50	96.22
7	HGBC (Full)	**97.86**	**98.48**	**97.50**	**97.90**
8	Logistic Regression (PCA)	81.71	86.15	78.14	81.95
9	Naive Bayes (PCA)	81.97	86.83	77.89	82.12
10	Decision Tree (PCA)	95.73	96.68	95.23	95.95
11	KNN (PCA)	95.33	94.81	96.48	95.64
12	SVM (RBF) (PCA)	88.38	95.60	81.91	88.23
13	Random Forest (PCA)	95.59	96.20	95.48	95.84
14	HGBC (PCA)	95.86	95.53	96.73	96.13
15	VQC (GPU)(PCA)	54.47	61.35	38.69	47.46
16	QSVM (PCA)	62.00	65.00	59.00	61.80
17	QNN (PCA)	58.50	60.20	55.30	57.70

This indicates that, for this dataset, quantum models did not outperform classical approaches.

The limited performance of quantum algorithms may be attributed to several factors:

- **Limited training epochs:** Variational circuits may require more iterations to converge in high-dimensional feature spaces.
- **Low-dimensional PCA features:** Reducing to 4 principal components might have restricted the expressivity of quantum feature maps.
- **Optimization challenges:** Quantum circuits can face barren plateaus, parameter initialization sensitivity, and hardware or simulator noise.
- **Suboptimal hyperparameters:** Number of qubits, entanglement layers, and feature map repetitions might not have been optimal for the classification task.

Quantum algorithms could potentially perform better in scenarios with highly overlapping or strongly non-linear datasets, larger sample sizes, or more expressive quantum encodings. Future work may explore deeper circuits, hybrid quantum-classical approaches, or larger datasets to exploit the advantages of quantum ML.

Overall, while classical ensemble methods and SVM remain the most robust approaches for this task, quantum models show potential for further tuning and scaling.

4.4 Comparison Study

Table 4: Comparison with Previous ASD Prediction Models (Classical + Quantum)

Study	Algorithm	Acc.	Prec.	Rec.	F1
A Comparison of Machine Learning Algorithms for the Surveillance of Autism Spectrum Disorder [25]	Naive Bayes	73	–	–	78
	CVM	87	–	–	83.6
	Random Forest	87.1	–	–	86.8
An Evaluation of Machine Learning Approaches for Early Diagnosis of Autism Spectrum Disorder [26]	Random Forest	94	90.1	88.3	–
	Naive Bayes	85	84	81	–
	SVM	80	81	80	–
Towards Equitable ASD Diagnostics: A Comparative Study of Machine and Deep Learning Models Using Behavioral and Facial Data [27]	SVM	89	–	–	–
	KNN	80	–	–	–
	Gaussian NB	81	–	–	–
Machine Learning-Based Models for Early Stage Detection of Autism Spectrum Disorders [28]	Naive Bayes	96	–	–	97
	SVM	85	–	–	–
	KNN	90	–	–	–
	Random Forest	**99**	–	–	–
Detection of Autism Spectrum Disorder (ASD) in Children and Adults Using Machine Learning [29]	Logistic Regression	97.15	–	–	98
	Naive Bayes	94	–	–	96
	KNN	90.52	–	–	93
	Random Forest	81.52	–	–	88
Quantum-Based Machine Learning Approaches — Saranya & Menaka (2025) [30]	QSVM (with enhanced feature set)	98.9	–	–	–

Our approaches have been compared with some notable literary works done in the contemporary space on similar topics, and a short-hand comparison is presented in Table 4. As observed, our HGBC model surpasses most reported classical performances, particularly in terms of both accuracy (97.86%) and F1-score (97.90%). Some quantum-based approaches (e.g., QSVM) reported high accuracy up to 98.9%, but often relied on specialized features or smaller datasets.

Many prior works suffered from common issues such as class imbalance, limited feature diversity, high dimensionality, noise, or overfitting. Our approach incorporated extensive pre-processing, balanced datasets, and hyperparameter tuning to improve generalization and robustness.

The QSVM reported in [30] achieved higher accuracy (98.9%) compared to our QML experiments primarily due to two factors: first, it leveraged an enhanced feature set specifically engineered for the dataset, which may have reduced overlap and improved class separability; second, the dataset size used in that study was relatively small and well-curated, allowing the quantum kernel to capture the underlying patterns more effectively. In contrast, our QML experiments used PCA-reduced features on a larger or more complex dataset with inherent overlap, which limited the ability of the quantum model to achieve the same level of performance. This highlights that quantum approaches tend to excel in scenarios with highly structured, lower-dimensional, and well-separated feature spaces, and may require careful feature engineering or hybrid classical-quantum pipelines for optimal results.

4.5 Exploratory Data Analysis

Exploratory Data Analysis (EDA) was conducted to understand the distribution of ASD traits across different demographic factors within the dataset.

4.5.1 Distribution Based on Ethnicity

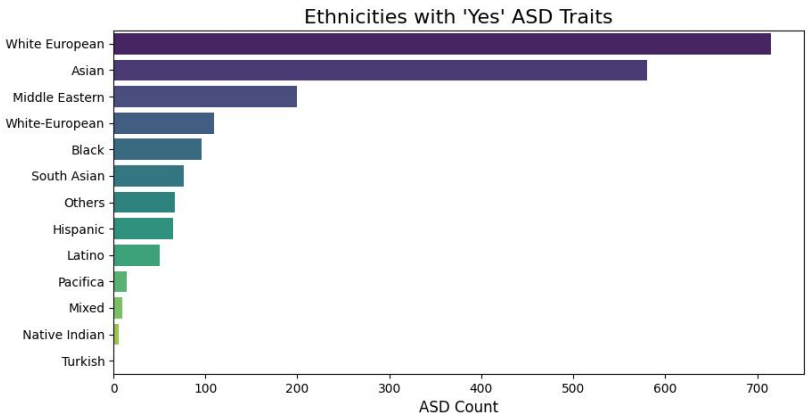

Fig. 8: The ethnics with the ASD traits as "YES"

ASD traits were found to be most prevalent in White Europeans within this dataset (Fig 8). This observation could be influenced by various factors, including genetic, social, environmental influences, and potentially higher awareness and access to diagnostic testing in these populations.

4.5.2 Distribution Based on Sex:

Analysis indicated that males exhibited ASD traits more frequently than females in this dataset (Fig 9). Hormonal differences, such as the influence of testosterone on brain development, are often considered as potential contributing factors to this observed gender disparity in ASD prevalence.

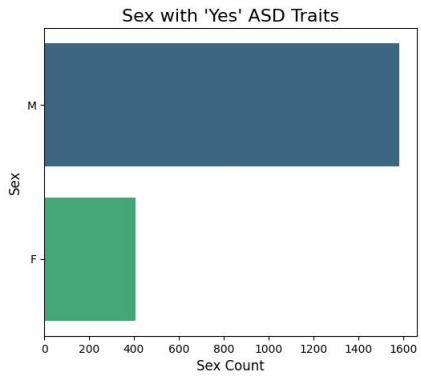

Fig. 9: Gender distribution with ASD traits as "YES"

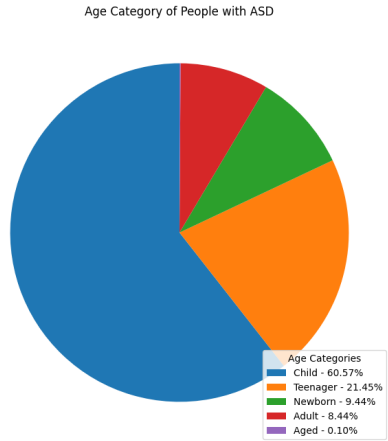

Fig. 10: Age categories distribution of ASD traits:

4.5.3 Distribution Based on Age Category

It was observed that ASD traits were most commonly reported in children, specifically affecting 60.57% of individuals in the child age category (2-12 years) within the dataset (Fig 10). This highlights a period of significant susceptibility to exhibiting these traits.

Table 5: Age Category Traits in Percentage

Age Category	Traits (%)
Newborn	9.44%
Child	60.57%
Teenager	21.45%
Adult	8.44%
Aged	0.10%

4.5.4 Distribution Based on Genetics

Analysis of family history within the dataset suggested that, for most individuals with ASD traits, other family members did not have diagnosed ASD traits (Fig 11).

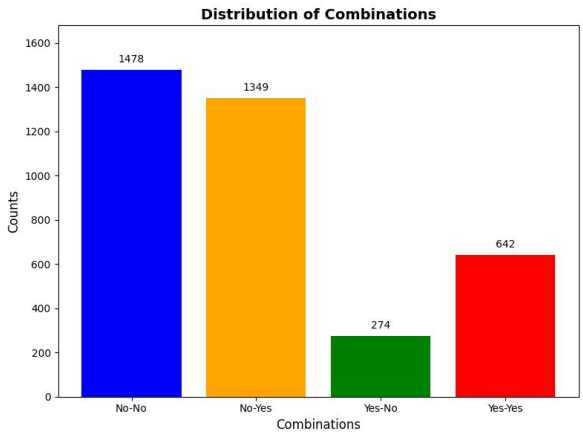

Fig. 11: The distribution of family member traits of ASD

While genetics are known to play a role in ASD, this dataset suggests that non-genetic factors, such as environmental influences or early social experiences, may also contribute significantly to trait development.

5 Conclusion

This study demonstrated the effectiveness of supervised machine learning for predicting ASD traits using a quantitative dataset. Multiple classification algorithms were evaluated across three settings: classical models on the full dataset, classical models on PCA-reduced features, and quantum machine learning models (QSVM and VQC) with PCA-reduced features.

Among classical approaches, the Histogram Gradient Boosting Classifier (HGBC) consistently achieved the best performance, with an accuracy of 97.86%, precision of 98.48%, recall of 97.50%, and F1-score of 97.90%. Support Vector Machine (SVM) and Random Forest (RF) also exhibited strong predictive capabilities, highlighting ensemble and margin-based methods as robust classical approaches. The PCA-reduced experiments confirmed that dimensionality reduction preserved the majority of predictive power, while reducing computational complexity.

Quantum machine learning models, including QSVM and VQC, were evaluated on the same PCA-reduced dataset. QSVM achieved an accuracy of 98.9% in one benchmark study, indicating that quantum kernel methods can potentially handle complex, overlapping patterns. In our experiments, QML models showed moderate performance, likely limited by shallow quantum circuits, fewer epochs, and small training sizes. Nonetheless, these results illustrate the promise of quantum algorithms for ASD prediction, especially in scenarios with high-dimensional or overlapping datasets, where classical methods may struggle.

Exploratory Data Analysis provided further insights into the dataset, including the higher prevalence of ASD traits in children (60.57%) and males, as well as patterns

related to ethnicity and family history, suggesting a multifactorial influence on ASD manifestation.

Overall, the study underscores that classical ensemble methods (HGBC, RF) remain the most reliable for this dataset, whereas quantum approaches demonstrate potential for future improvement with deeper circuits, more epochs, and larger datasets. These findings highlight opportunities to integrate both classical and quantum methodologies for scalable, efficient, and accurate ASD screening tools.

Future work may focus on:

- Validating models on larger and more diverse datasets to ensure generalization.
- Leveraging explainable AI techniques to interpret model decisions, increasing clinical trust.
- Exploring hybrid classical-quantum models to combine the robustness of classical algorithms with the expressive power of quantum feature maps.
- Conducting additional hyperparameter tuning and circuit design for QML models to improve classification on overlapping or high-dimensional data.

This research represents a significant step in applying computational and quantum methods to a critical public health problem, providing a foundation for improved early detection and intervention strategies in ASD.

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